

# AIR PRESSURE

## — CAN CRUSH! —

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### Grade 2 Science Lesson

*Discover how air pushes, presses & crushes!*

■ Subject: Science

■ Grade: 2

■ Duration: 45 mins

■ Type: Hands-on Experiment

PLANRS ACADEMY

## ■ What is Air Pressure?

Air (Hawa / ■■■■) is all around us — and it is always **pushing!** This push is called **Air Pressure**. Air pushes in every direction: up, down, left, right, and from all sides at once.

## ■ Air Is Everywhere and Always Pushing

Imagine holding a balloon. Air *inside* the balloon pushes the rubber outwards in every direction — that is why a balloon is round! The air *outside* also pushes inward, which keeps the balloon balanced.

■ **KEY IDEA:** Air pushes outward from inside a balloon. Outside air pushes inward. When balanced, the balloon keeps its shape!

## ■ Think & Predict!

### Question for students:

What do you think will happen if we take **all the air OUT** of a plastic bottle and then seal it tightly?

Make your guess below ■

A) It stays the same

B) It gets bigger

**C) It gets crushed! ■**

D) It floats away

## ■ The Bottle-Crush Experiment

### ■ Materials You Need:

■ Your lungs

■ An empty plastic bottle

■ A bottle cap (lid)

### ■ Follow These Steps:

**1**

Take an empty plastic bottle.

**2**

Put the bottle to your mouth and **suck all the air out**.

**3**

While still sucking, quickly **put on the cap** to seal it.

**4**

Take the bottle away from your mouth and **watch what happens!**

## ■ What Did We See?

The bottle gets **crushed and squashed** — even though nobody touched it! The bottle stayed squashed even after we let go.

**Why?** Because we sucked the air OUT, there was no air inside to push outward. The outside air (Hawa) continued pushing **inward** (Dhakka / ■■■■■■) from all sides, squeezing (Dabna / ■■■■■) the bottle flat!

## ■ The Science Behind It

| Inside the bottle              | Outside the bottle              |
|--------------------------------|---------------------------------|
| No air → No outward push       | Normal air → Strong inward push |
| Pressure = LOW                 | Pressure = HIGH                 |
| Result: bottle walls cave IN ■ | Outside wins → CRUSH! ■         |

■ The air around us is always pressing. When inside pressure drops, outside air pressure wins and crushes the bottle!

## ■ Real Life Examples of Air Pressure

### Real Life Example 1: Pressure Cooker Whistle ■

The whistle on a pressure cooker lets out extra pressure (Dabaav / ■■■■) so the cooker does not explode. The steam builds pressure inside — the whistle is the safety valve that releases it and lets food cook safely.

### Real Life Example 2: Biscuit Packet Pop! ■

When you open a biscuit packet at high altitude (like on a mountain), the air inside the packet is at higher pressure than the air outside. It pops open because the inside air pushes outward — just like our bottle, but in reverse!

### Real Life Example 3: Steel Tiffin Dabba ■

A metal tiffin box stays strong because air pressure inside and outside is **balanced** (Santulan / ■■■■■■). Both sides push with equal force, so the box does not crush or expand.

### Real Life Example 4: Ears Popping on Mountain Trips ■■

When you travel up a mountain, the air outside gets thinner — less air pressure. The air inside your ears is still at higher pressure, so it pushes outward and makes your ears **pop**!

## ■ Vocabulary: Important Words to Know

### Pressure

The pushing force of air on surfaces (Dabaav)

### Air (Hawa)

The invisible gas all around us (■■■■)

### Crush

To squeeze and squash by force (Dabna / ■■■■)

### Push

To press against something (Dhakka / ■■■■■)

### Balance

Equal pushing on both sides (Santulan / ■■■■■■)

## ⇒ ■ Quick Quiz — Check What You Learned!

**Q1. What happens when we take air OUT of a sealed bottle?**

A) It stays the same B) It gets crushed ■ C) It gets bigger D) It floats

**Q2. What is the pushing force of air called?**

A) Weight B) Gravity C) Air Pressure ■ D) Wind

**Q3. Why does a pressure cooker whistle?**

A) To make noise B) To release extra pressure ■ C) To cool down D) To add air

**Q4. What does 'Santulan' mean?**

A) Crush B) Push C) Balance ■ D) Air

## ■ Lesson Summary

■ Air (Hawa) is all around us and always pushing.

■ This pushing force is called **Air Pressure**.

■ When air is removed from inside a bottle, outside pressure **crushes** it.

■ Air pressure is balanced (Santulan) when inside and outside pressure are equal.

■ We see air pressure in everyday life: pressure cookers, biscuit packets, ears popping!

### ■ Coming Up Next!

Great job exploring air pressure today! Keep watching for our next lesson: **How Does Air Help Balloons Fly?**

You'll discover how air helps things move, lift, and soar into the sky! ■

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**You Are a Science Detective! ■ Great job exploring today!**